25901282 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECFSC200 - GOOD LABORATORY PRACTICES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Explain the role of NABL in maintaining laboratory standards in India. [K] / [CO1]
2. State the importance of equipment calibration in testing laboratories. [U] / [CO2]
3. Identify two situations where fire extinguishers must be used immediately. [U] / [CO3]
4. Apply standard buffer calibration procedure for pH meter. [A] / [CO4]
5. Expand ISO. [U] / [CO1]
6. Estimate the risk level in a laboratory storing flammable solven [E] / [CO2]
7. Apply competency assessment strategies in laboratory settings. [A] / [CO3]
8. Analyze importance of SOP adherence in maintaining consistency. [An] / [CO4]
9. List any two essential infrastructure requirements for maintaining quality in a forensic laboratory. [K] / [CO2]
10. Examine the impact of improper storage of volatile chemicals. [An] / [CO2]

11. Explain the role of breakdown service documentation in quality control. [U] / [CO3]

12. List any two types of chemical protective clothing. [K] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the significance of corrective action in quality management. [U] / [CO1]

14. Describe the system of document control in a laboratory quality management system. [U] / [CO2]

15. Explain the importance of maintaining personnel training and competency records. [U] / [CO2]

16. Analyze measurement uncertainty due to calibration errors. [An] / [CO4]

17. Explain how internal audits contribute to continuous quality improvement. [U] / [CO1]

18. Discuss the safety precautions to be followed while handling hazardous chemicals. [A] / [CO2]

19. Explain the purpose and functions of ISO in quality management. [U] / [CO1]

[4x5 = 20]

Part C

Essay questions

Answer any **1** questions

Each question carries **10** marks

20. Describe the standard safety precautions and risk assessment procedures required while handling, analyzing, and storing hazardous chemical reagents in a laboratory. [S] / [CO2]

21. Analyze role of documentation in preventing quality failures. [An] / [CO4]

[10x1 = 10]